

Statistical Programming Languages

R Markdown reports

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Why R Markdown?

- ▶ Help us create a document that embeds both text and code.
- ▶ Advantages:
 - ▶ Research becomes reproducible.
 - ▶ Makes it easier to share analysis results together with their documentation.
 - ▶ When you “knit” the document, code is executed and the output appears alongside the text.
 - ▶ The document renders only if there are no errors (warnings can be shown or suppressed).
 - ▶ You can render the document to HTML, PDF, or Word.
- ▶ Installation if you are not using RStudio
 - ▶ `pandoc` your swiss-army knife to convert files from one markup format into another

```
install.packages("tinytex")    # TeX for generating PDF
install.packages("rmarkdown")
```

New R Markdown file - RStudio

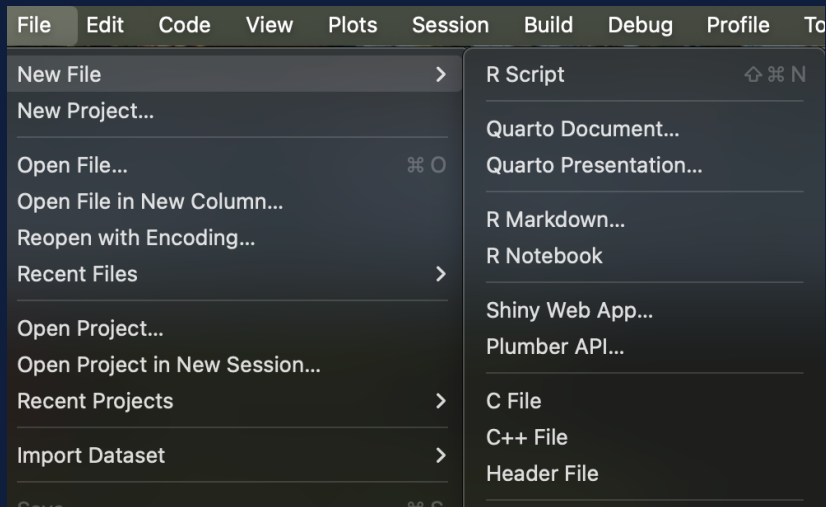


Figure 1: Create a new R Markdown file

New R Markdown file - Dialogbox

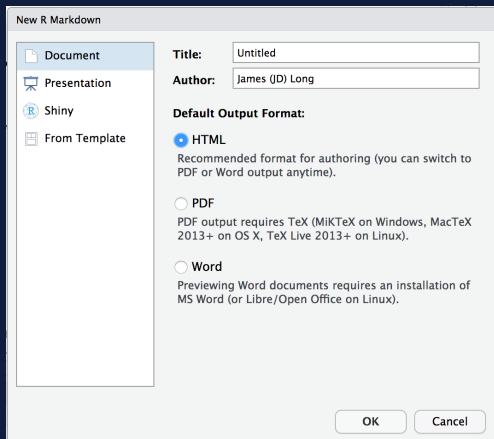


Figure 2: Dialogbox for new R Markdown file

New R Markdown file - Default file

title: "Untitled"

author: Katarzyna Reluga

output: pdf_document

date: "2025-09-17"

```
```{r setup, include=FALSE}
```

```
knitr::opts_chunk$set(echo = TRUE)
```

```
```
```

R Markdown

This is an R Markdown document.

Markdown is a simple formatting syntax for authoring HTML,

```
```{r cars}
```

```
summary(cars)
```

```
```
```

R Markdown file structure

- ▶ YAML — a human-readable **data format** (not markup).
Name: “YAML Ain’t Markup Language” (originally “Yet Another Markup Language”).
- ▶ Markdown
 - ▶ defined by Gruber and Swartz (2004)
 - ▶ lightweight markup language with plain-text-formatting syntax
 - ▶ can be converted to many output formats
- ▶ R Markdown
 - ▶ integration of R code into markdown documents
- ▶ [R Markdown Cheat Sheet](#)

```
---
```

```
YAML header
```

```
---
```

```
R Markdown with code chunks
```

YAML excerpt

| Element | Meaning |
|--------------------------|---|
| --- | start block |
| # | comment start (till end of line) |
| - xxx | element of item list |
| [xxx, yyy, zzz] | inline item list |
| key: value | element of named item list |
| {key1: val1, key2: val2} | named inline item list |
| key: | multiline string preserving new lines |
| key: > | multiline string empty line = new paragraph |

► [yaml](#): methods to Convert R Data to YAML and Back

YAML - The example header

```
library("yaml")
res <- read_yaml(text='
---
title: "Untitled"
author: "Katarzyna Reluga"
date: "4 March 2025"
output: pdf_document
---')
str(res)
## List of 4
## $ title : chr "Untitled"
## $ author: chr "Katarzyna Reluga"
## $ date  : chr "4 March 2025"
## $ output: chr "pdf_document"
```


YAML header for vignette I

```
---  
# Title shown on the rendered document  
title: "gettext"  
  
# Author block (can list multiple authors)  
author:  
  - name: "Katarzyna Reluga"          # displayed author name  
    email: "katarzyna.reluga@hu-berlin.de"  # contact  
  
# Date: inline R code evaluates at knit time (e.g., "17 September, 2025")  
date: "`r format(Sys.time(), '%d %B, %Y')`"  
  
# Output format and options (here: HTML with a table of contents)  
output:  
  html_document:  
    toc: true  
  
# R package vignette metadata (passed through to LaTeX/Pandoc)  
# '>' = folded block; the lines below are literal content, not YAML keys  
vignette: >  
  %\VignetteIndexEntry{gettext}  
  %\VignetteEngine{knitr::rmarkdown}  
  \usepackage[utf8]{inputenc}  
---
```

YAML header for vignette II

```
res <- read_yaml("gettext.Rmd")
str(res)
## List of 5
## $ title : chr "gettext"
## $ author :List of 1
## ..$ :List of 2
## .. ..$ name : chr "Katarzyna Reluga"
## .. ..$ email: chr "katarzyna.reluga@hu-berlin.de"
## $ date : chr "`r format(Sys.time(), '%d %B, %Y')`"
## $ output :List of 1
## ..$ html_document:List of 1
## .. ..$ toc: logi TRUE
## $ vignette: chr "%\\VignetteIndexEntry{gettext} %\\VignetteEngine{k
```

Some output formats

| Value | Format |
|--------------------------------------|-------------------------------------|
| <code>html_notebook</code> | Interactive R Notebooks |
| <code>html_document</code> | HTML document w/ Bootstrap CSS |
| <code>pdf_document</code> | PDF document (via LaTeX template) |
| <code>word_document</code> | Microsoft Word document (docx) |
| <code>odt_document</code> | Open Document Text document |
| <code>rtf_document</code> | Rich Text Format document |
| <code>md_document</code> | Markdown document (various flavors) |
| <code>beamer_presentation</code> | PDF presentation with LaTeX Beamer |
| <code>powerpoint_presentation</code> | PowerPoint presentation |
| <code>html_vignette</code> | R package vignette (HTML) |
| <code>github_document</code> | GitHub Flavored Markdown document |

- ▶ Specific options (e.g. `toc`) depend on the output format
- ▶ for details see [R Markdown: The Definitive Guide, Chapter 3 ff.](#)

Some render option for PDF I

- ▶ PDF document: $\text{Rmd} \xrightarrow{\text{knitr}} \text{md} \xrightarrow{\text{pandoc}} \text{tex} \xrightarrow{\text{pdflatex}} \text{pdf}$
- ▶ Beamer is a $\text{\LaTeX}$ document class for creating presentation slides

| Option (only PDF) | Description |
|------------------------------|--|
| <code>number_sections</code> | Add section numbering to headers |
| <code>toc_depth</code> | The lowest level of headings to add to table of contents |

| Option (only beamer) | Description |
|--------------------------|--|
| <code>colortheme</code> | Beamer color theme to use |
| <code>incremental</code> | Should bullets appear one at a time (on presenter mouse clicks)? |
| <code>slide_level</code> | The lowest heading level that defines individual slides |
| <code>theme</code> | Bootswatch or Beamer theme to use for page |

Some render option for PDF II

| Option | Description |
|---|--|
| <code>citation_package</code> | The LaTeX package to process citations, <code>natbib</code> , <code>biblatex</code> or <code>none</code> |
| <code>dev</code> | Graphics device to use for figure output (e.g. "png") |
| <code>fig_caption</code> | Should figures be rendered with captions? |
| <code>fig_height</code> ,
<code>fig_width</code> | Default figure height and width (in inches) for document |
| <code>highlight</code> | Syntax highlighting |
| <code>includes</code> | File of content to place in document (<code>in_header</code> , <code>before_body</code> , <code>after_body</code>) |
| <code>keep_md</code> | Save a copy of <code>.md</code> file that contains knitr output |
| <code>keep_tex</code> | Save a copy of <code>.tex</code> file that contains knitr output |
| <code>latex_engine</code> | Engine to render latex |
| <code>md_extensions</code> | Markdown extensions to add to default definition or R Markdown |
| <code>pandoc_args</code> | Additional arguments to pass to Pandoc |
| <code>template</code> | Pandoc template to use when rendering file |
| <code>toc</code> | Add a table of contents at start of document |

Example – slides for today

```
---
title: "Statistical Programming Languages"
subtitle: "R Markdown reports"
author:
  - "Katarzyna Reluga\\thanks{Acknowledgements: Thanks to Manuel Pfeuffer, Maar
date: ""
output:
  beamer_presentation:
    keep_tex: true
    highlight: zenburn
header-includes:
  - \usepackage{myhu-darktheme} # load my customized theme
  - \setbeamertemplate{navigation symbols}{} # hide Beamer's nav buttons
  - \setbeamertemplate{footline}[frame number] # show slide number
colorlinks: true
linkcolor: cyan
urlcolor: cyan
citecolor: cyan
---
```

R Markdown - text formatting

| Syntax | Result |
|--|---|
| Plain text | Plain text |
| <code>*italics*, _italics_</code> | <i>italics, italics</i> |
| <code>**bold**, __bold__</code> | bold, bold |
| <code>superscript^2^</code> | superscript ² |
| <code>~~strikethrough~~</code> | strikethrough |
| <code>`verbatim code`</code> | verbatim code |
| <code><!-- text comment --></code> | |
| <code>http://www.rstudio.com</code> | http://www.rstudio.com |
| <code>[RS] (http://www.rstudio.com)</code> | RS |
| <code>! [Caption] (Rlogo.png)</code> |  |

R Markdown text - Headers

| Syntax | Result |
|-----------------------------|----------|
| <code># Header 1</code> | Header 1 |
| <code>## Header 2</code> | Header 2 |
| <code>### Header 3</code> | Header 3 |
| <code>#### Header 4</code> | Header 4 |
| <code>##### Header 5</code> | Header 5 |
| <code>##### Header 6</code> | Header 6 |

- ▶ `# Header 1` is often used for the title header
- ▶ start with `## Header 2`, e.g. in vignettes

R Markdown text - Unordered lists

```
* unordered list
  + sub-item 1
  + sub-item 2
    - sub-sub-item 1
* item 2
```

- ▶ unordered list
 - ▶ sub-item 1
 - ▶ sub-item 2
 - ▶ sub-sub-item 1
- ▶ item 2

R Markdown text - Ordered and definition lists

```
1. ordered list
2. item 2
    i) sub-item 1
        A. sub-sub-item 1
```

```
1. ordered list
2. item 2
    i) sub-item 1
        A. sub-sub-item 1
```

```
Term 1
: Definition 1
```

Term 1 Definition 1

R Markdown text - Table

```
Right	Left	Default	Center
12	12	12	12
123	123	123	123
1	1	1	1
```

| Right | Left | Default | Center |
|-------|------|---------|--------|
| 12 | 12 | 12 | 12 |
| 123 | 123 | 123 | 123 |
| 1 | 1 | 1 | 1 |

- ▶ the positions of the : determines the alignment of a column
- ▶ the number of - determines the width of each column

R Markdown text - Math formulas

- ▶ Math formulas can be done with $\LaTeX$
- ▶ Euler identity as formula: $e^{i\pi} + 1 = 0$

$$e^{i\pi} + 1 = 0$$

- ▶ Euler identity as inline: The Euler identity $e^{i\pi} + 1 = 0$ is beautiful.

The Euler identity $e^{i\pi} + 1 = 0$ is beautiful.

- ▶ [\$\LaTeX\$ math cheat sheet](#)
- ▶ Hint: If you know $\LaTeX$ (e.g. `\LaTeX{}`) then you can embed $\LaTeX$ commands in R Markdown text
 - ▶ This works properly when generating PDFs, but not for HTML

R Markdown text - Slide break & footnotes

horizontal line (PDF) / slide break (beamer):

```
***
```

```
A footnote [^1] and a remark about them[^2]
```

```
[^1]: Here is the footnote
```

```
[^2]: In general, the use of footnotes in most  
academic writing is now considered a bit outdated and  
has been replaced by endnotes.
```

A footnote ² and a remark about them³

²Here is the footnote

³In general, the use of footnotes in most academic writing is now considered a bit outdated and has been replaced by end notes.

R Markdown - Text output

► Default:

```
```{r}
"A simple R program"
```

"A simple R program"
## [1] "A simple R program"
```

► No echoing

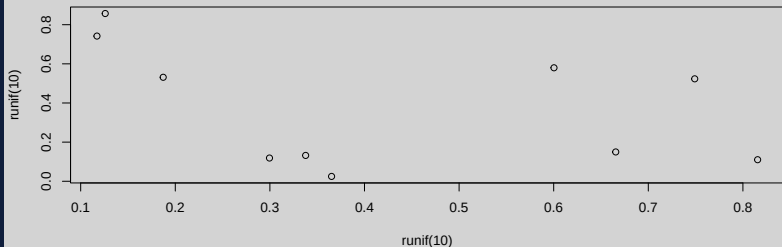
```
```{r, echo=FALSE}
"A simple R program"
```

## [1] "A simple R program"
```

R Markdown - Plot

```
```{r, fig.height=4}  
par(bg = "lightgrey")
plot(runif(10), runif(10))
```
```

```
par(bg = "lightgrey")  
plot(runif(10), runif(10))
```



R Markdown text - Tables from R

```
tab <- table(iris$Species)
knitr::kable(tab)
```

| Var1 | Freq |
|------------|------|
| setosa | 50 |
| versicolor | 50 |
| virginica | 50 |

```
knitr::kable(HairEyeColor[,,'Female'])
```

| | Brown | Blue | Hazel | Green |
|-------|-------|------|-------|-------|
| Black | 36 | 9 | 5 | 2 |
| Brown | 66 | 34 | 29 | 14 |
| Red | 16 | 7 | 7 | 7 |
| Blond | 4 | 64 | 5 | 8 |

R Markdown - Some chunk options

| Option | Default | Description |
|-------------------------|-----------------------|--|
| <code>child</code> | <code>NULL</code> | file(s) to knit and then include |
| <code>collapse</code> | <code>FALSE</code> | collapse all output into single block |
| <code>comment</code> | <code>'##'</code> | prefix for each line of results |
| <code>dependson</code> | <code>NULL</code> | chunk dependencies for caching |
| <code>echo</code> | <code>TRUE</code> | Display code in output document |
| <code>error</code> | <code>TRUE</code> | Display error messages in doc(<code>TRUE</code>) or stop render when errors occur (<code>FALSE</code>) |
| <code>eval</code> | <code>TRUE</code> | Run code in chunk |
| <code>fig.align</code> | <code>"center"</code> | figure alignment |
| <code>fig.cap</code> | <code>NULL</code> | figure caption as character string |
| <code>fig.height</code> | <code>7</code> | Height of plots in inches |
| <code>fig.width</code> | <code>7</code> | Width of plots in inches |
| <code>highlight</code> | <code>TRUE</code> | highlight source code |
| <code>include</code> | <code>TRUE</code> | Include chunk in doc after running |
| <code>message</code> | <code>TRUE</code> | display code messages in document |
| <code>results</code> | <code>"markup"</code> | <code>"asis"</code> pass through results, <code>"hide"</code> do not display results, <code>"hold"</code> put all results below all code |
| <code>tidy</code> | <code>FALSE</code> | tidy code for display |
| <code>warning</code> | <code>TRUE</code> | display code warnings in document |
